

Active Learning Performance Curves

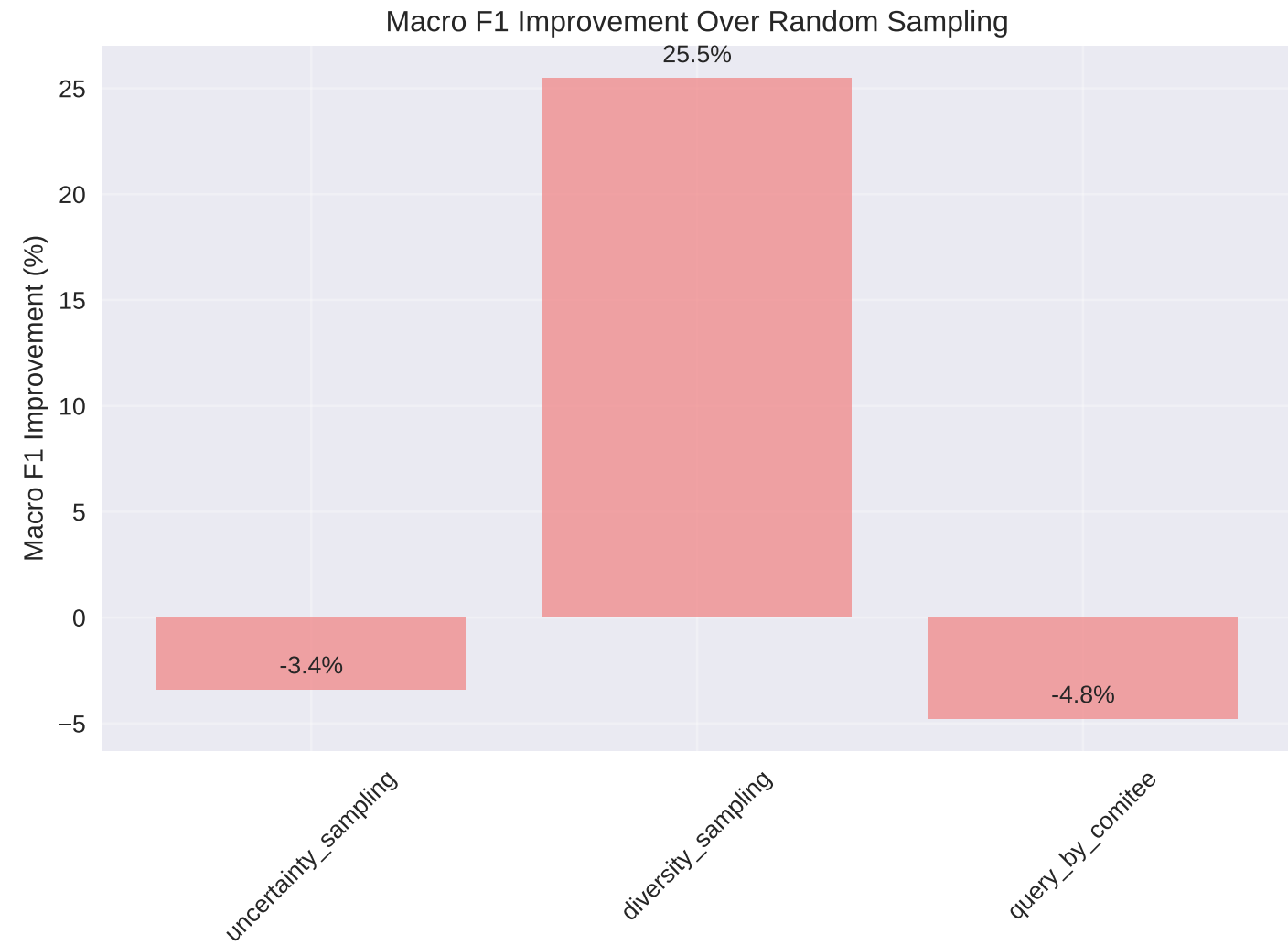
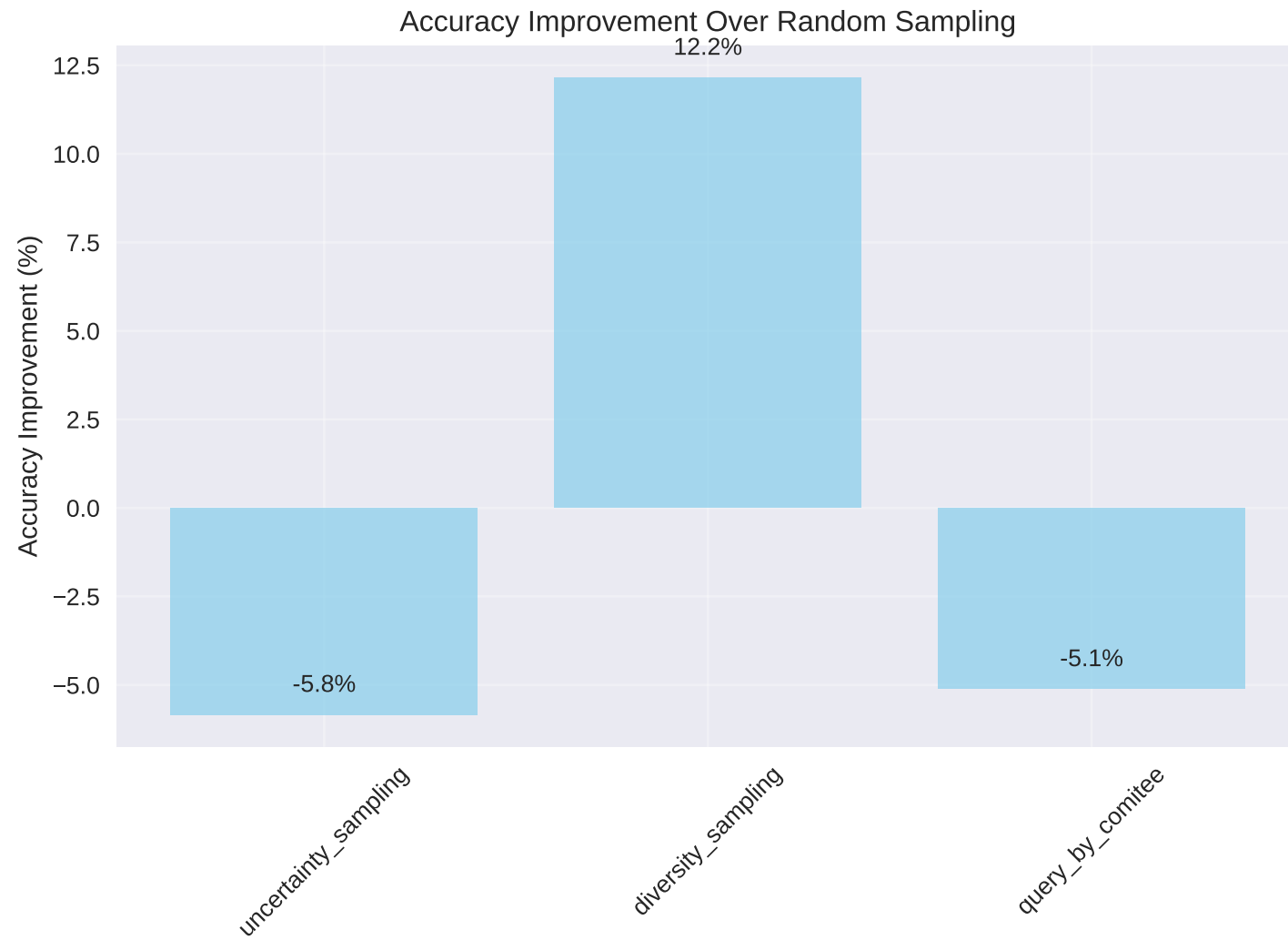
Accuracy vs Training Data Size



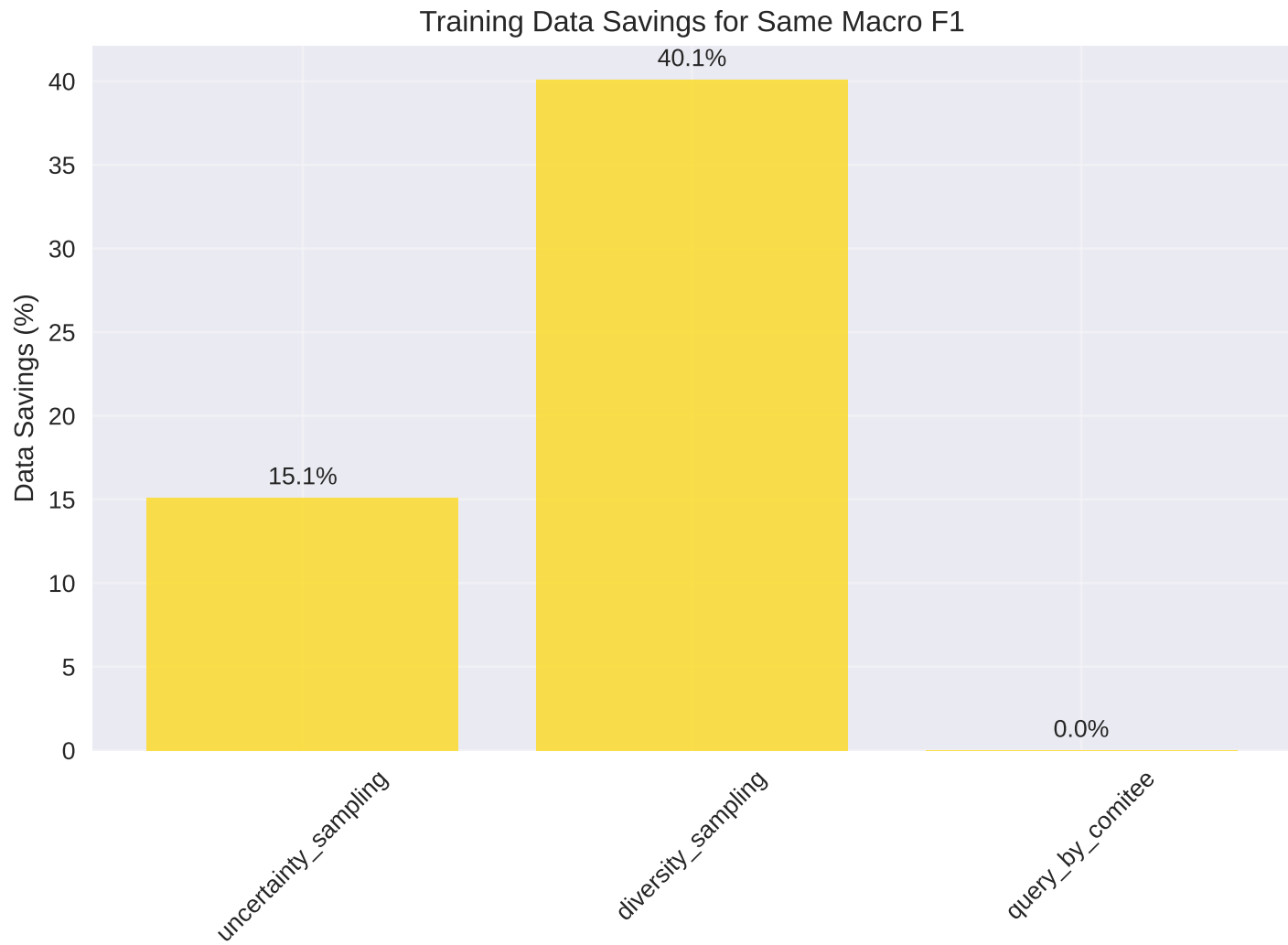
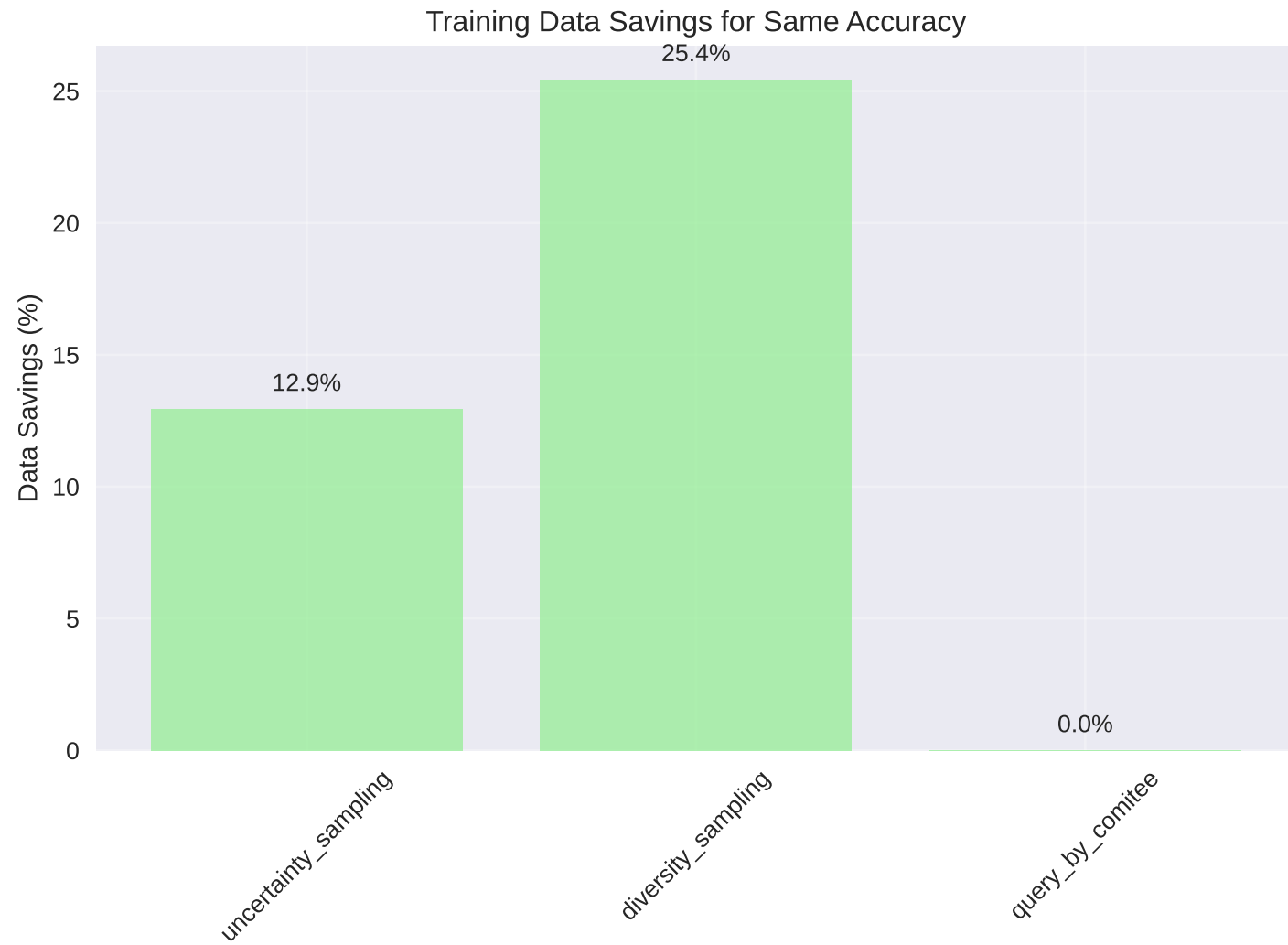
Macro F1 vs Training Data Size



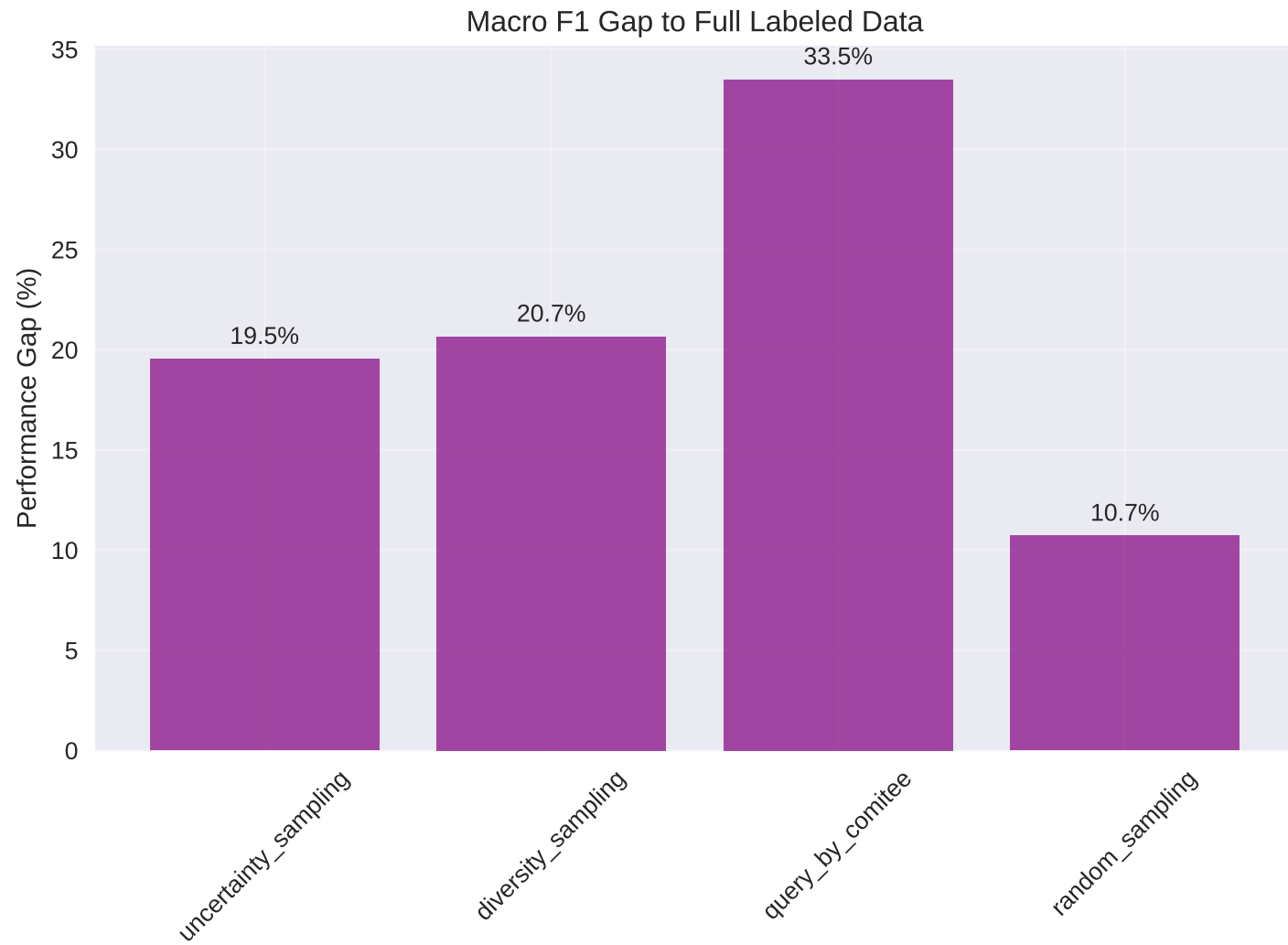
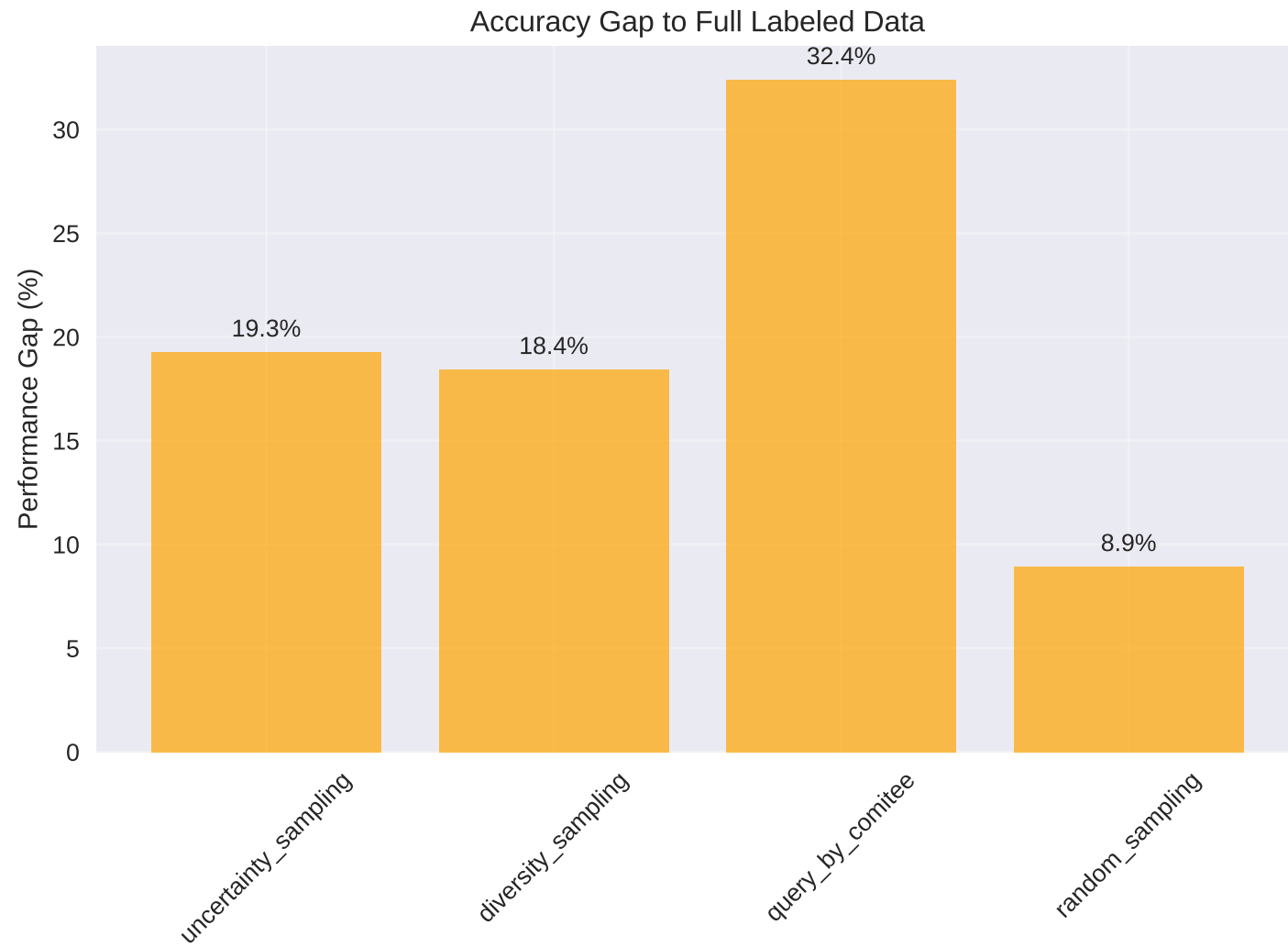
Performance Improvements vs Random Sampling



Data Efficiency Analysis



Performance Gap to Full Labeled Data



FULL LABELED DATA REFERENCE:

- Accuracy: 0.716
- Macro F1: 0.725
- Training Size: 1000

PERFORMANCE IMPROVEMENT OVER RANDOM SAMPLING:

Uncertainty Sampling:

- Accuracy Improvement: -5.84%
- Macro F1 Improvement: -3.38%
- Max Accuracy Improvement: 94.52%
- Max Macro F1 Improvement: 223.61%

Diversity Sampling:

- Accuracy Improvement: 12.16%
- Macro F1 Improvement: 25.49%
- Max Accuracy Improvement: 293.15%
- Max Macro F1 Improvement: 574.13%

Query By Comitee:

- Accuracy Improvement: -5.11%
- Macro F1 Improvement: -4.76%
- Max Accuracy Improvement: 167.12%
- Max Macro F1 Improvement: 290.56%

DATA EFFICIENCY ANALYSIS:

Uncertainty Sampling:

- Avg Accuracy Data Savings: 12.94%
- Avg Macro F1 Data Savings: 15.10%
- Max Accuracy Data Savings: 72.22%
- Max Macro F1 Data Savings: 93.75%

Diversity Sampling:

- Avg Accuracy Data Savings: 25.44%
- Avg Macro F1 Data Savings: 40.10%
- Max Accuracy Data Savings: 72.22%
- Max Macro F1 Data Savings: 93.75%

Query By Comitee:

- Avg Accuracy Data Savings: 0.00%
- Avg Macro F1 Data Savings: 0.00%
- Max Accuracy Data Savings: 0.00%
- Max Macro F1 Data Savings: 0.00%

COMPARISON WITH FULL LABELED DATA:

Uncertainty Sampling:

- Final Accuracy: 0.578
- Final Macro F1: 0.583
- Final Training Size: 400
- Accuracy Gap to Full: 19.27%
- Macro F1 Gap to Full: 19.54%
- Data Efficiency vs Full: 40.00%

Diversity Sampling:

- Final Accuracy: 0.584
- Final Macro F1: 0.575
- Final Training Size: 300
- Accuracy Gap to Full: 18.44%
- Macro F1 Gap to Full: 20.65%
- Data Efficiency vs Full: 30.00%

Query By Comitee:

- Final Accuracy: 0.484
- Final Macro F1: 0.482
- Final Training Size: 300
- Accuracy Gap to Full: 32.40%
- Macro F1 Gap to Full: 33.49%
- Data Efficiency vs Full: 30.00%

Random Sampling:

- Final Accuracy: 0.652
- Final Macro F1: 0.647
- Final Training Size: 1900
- Accuracy Gap to Full: 8.94%
- Macro F1 Gap to Full: 10.71%
- Data Efficiency vs Full: 190.00%

STATISTICAL ANALYSIS:

Methods Analyzed: uncertainty_sampling, diversity_sampling, query_by_comitee, random_sampling

Accuracy Statistics:

- Mean: 0.575
- Std Dev: 0.060
- Min: 0.484
- Max: 0.652

Macro F1 Statistics:

- Mean: 0.572
- Std Dev: 0.059
- Min: 0.482
- Max: 0.647

Uncertainty Sampling vs Random Sampling:

- Accuracy Difference: -0.074
- Macro F1 Difference: -0.064
- Accuracy Improvement: -11.35%
- Macro F1 Improvement: -9.89%

Diversity Sampling vs Random Sampling:

- Accuracy Difference: -0.068
- Macro F1 Difference: -0.072
- Accuracy Improvement: -10.43%
- Macro F1 Improvement: -11.14%

Query By Comitee vs Random Sampling:

- Accuracy Difference: -0.168
- Macro F1 Difference: -0.165
- Accuracy Improvement: -25.77%
- Macro F1 Improvement: -25.51%

KEY FINDINGS:

- Best performing method: Diversity Sampling
- Highest accuracy improvement: 12.16% over random sampling
- Most data efficient: Diversity Sampling
- Highest data savings: 25.44% for same accuracy
- Closest to full labeled performance: Random Sampling
- Smallest performance gap: 8.94% to full data